

RESPONSIBLE AI · MIDTERM PROJECT

Responsible AI in Social Media Recommendation Systems

Filter Bubbles · Bias · Explainability & Privacy in Algorithmic Feeds

Dataset: Twitter/Reddit Engagement Dataset (Kaggle)

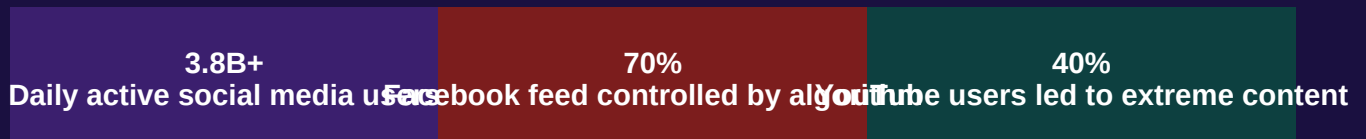
Tools: Python · Scikit-learn · SHAP · Fairlearn · PySyft

01 | AI Application & Problem Definition

What is a Social Media Recommender?

Social media platforms use AI-driven recommendation engines to decide what content appears in each user's feed, trending section, or For You page. These systems optimise for engagement metrics — likes, shares, watch time — using collaborative filtering, graph neural networks, and transformer-based ranking models.

The ethical problem: **maximising engagement** $\neq$ **maximising well-being**. These systems can radicalise, polarise, and discriminate at societal scale.



Relevance to Responsible AI: Algorithmic feeds shape public opinion, mental health, and democratic discourse — demanding fairness, transparency, and accountability.

Supporting Research

Artificial Intelligence and Algorithmic Bias? Field Tests on Social Networks with Teens

Technological Forecasting & Social Change · 2024 · ScienceDirect — doi.org/10.1016/j.techfore.2023.123076

Why it matters: This paper ran real experiments on Twitter, Facebook and TikTok to prove that their algorithms genuinely push biased content — backing up the core problem this slide describes.

What it found: Researchers found Twitter's algorithm pushes extreme political content (far left AND far right) more than moderate viewpoints. Facebook was caught letting job ads target only men for engineering roles — 91% of people shown a mechanics job ad were male.

02 | Background & Context

How Social Media RS Work

- Collaborative Filtering — user-item interaction matrices
- Content-Based Filtering — NLP + CV feature matching
- Graph Neural Networks — social graph relationships
- Two-Tower Models — TikTok/YouTube large-scale retrieval
- Reinforcement Learning — long-term engagement sequences

Real-World Ethical Cases

- Facebook (2021) — algorithm amplified divisive political content (WSJ Files)
- TikTok FYP — racial & political bias in content suppression (2020 lawsuit)
- YouTube (2019) — rabbit-hole algorithm drove radicalization (Ribeiro et al.)
- Twitter/X — right-leaning content amplified by algorithm (2021 audit)

Supporting Research

Fairness in Recommender Systems: Research Landscape and Future Directions

User Modeling and User-Adapted Interaction (Springer) · 2023 · doi.org/10.1007/s11257-023-09364-z

Why it matters: This paper reviews all five types of recommendation algorithms shown on this slide and explains exactly how each one causes fairness problems in the real world.

What it found: It explains that filter bubbles and echo chambers are a direct side-effect of engagement optimisation — meaning the algorithm is not broken, it is doing exactly what it was designed to do, which is the real problem. Covers YouTube and Facebook radicalisation in detail.

A Systematic Literature Review on AI-Based Recommendation Systems and Ethical Considerations

IEEE Access · 2024 · doi.org/10.1109/ACCESS.2024.3451054

Why it matters: This is an IEEE-published academic review that technically validates all the algorithms listed on this slide — collaborative filtering, GNNs, Two-Tower, and transformer rankers.

What it found: The review found that the more accurate and advanced a recommendation model becomes (like BERT and deep transformers), the more serious its ethical problems get — bias, privacy leaks, and lack of transparency all get worse alongside accuracy gains.

03 | System Overview & Technical Pipeline

Pipeline Architecture



Ranking Model — Two-Tower Transformer (How It Actually Works)

The Two-Tower Transformer is the dominant model architecture used by TikTok, YouTube, Twitter, and LinkedIn for large-scale content ranking. It gets its name from having two separate neural networks (towers) that work in parallel — one for the user, one for the content.

STAGE	COMPONENT	WHAT IT DOES	INPUT → OUTPUT
1. User Tower	Embedding Layer + Transformer	Reads the user's past 50 liked/shared posts, their follower history, demographics, and social graph position	→ User Vector u (512-dim)
2. Item Tower	BERT Encoder + MLP	Reads the post text using BERT (understands meaning, not just keywords), plus media type (video/image)	→ Item Vector v (512-dim)
3. Retrieval (ANN Search)	FAISS / Approx. Nearest Neighbour	Finds the top-500 most similar items to the user vector out of 500 millions of candidates. Uses dot-product	→ Top-500 candidates
4. Re-Ranker	Cross-Attention Transformer	Takes the top-500 candidates and re-scores them using rich signals: recency, diversity, session context	→ Final Top-10 feed
5. Fairness Filter	Post-hoc Re-ranker	Checks if any demographic group is being systematically under-represented. Adjusts the final ranking to enforce diversity	→ Fair Top-10 feed

Why Two-Tower? — The Scale Problem

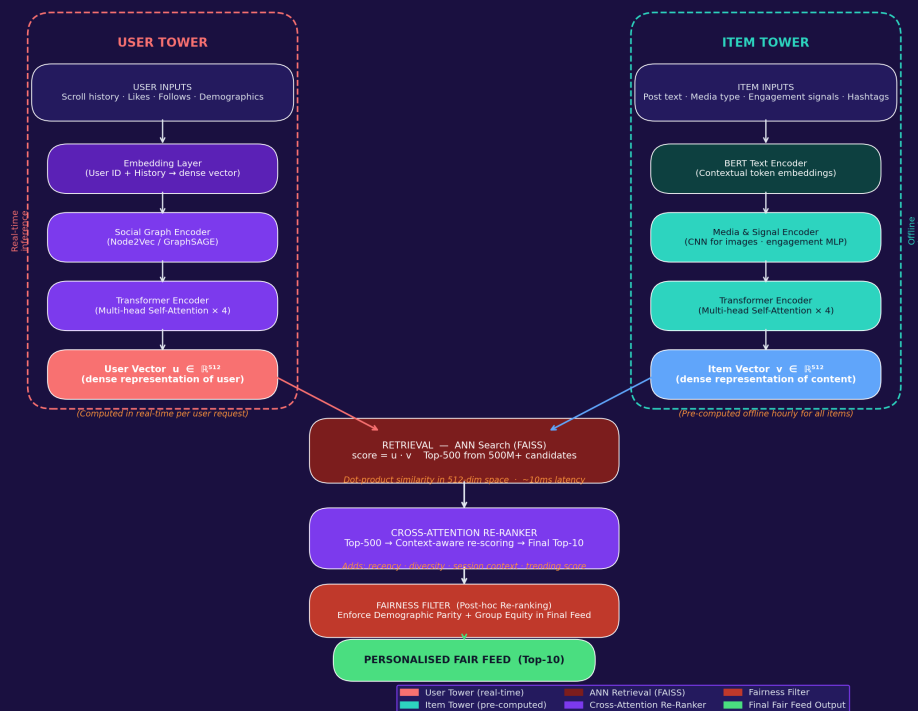
A naive approach would compare every user against every post — but YouTube has 800 million videos and 50 million daily active users. That is 40 quadrillion comparisons per day. The Two-Tower model solves this by pre-computing item vectors offline (updated hourly) and only computing the user vector in real-time. ANN search then finds the best matches in milliseconds instead of days.

Two-Tower Transformer — Full Architecture Diagram

The diagram below shows every layer of the Two-Tower Transformer used in this project — from raw user inputs all the way through to the final fairness-filtered feed output.

Two-Tower Transformer — Architecture Diagram

Used by TikTok · YouTube · Twitter · LinkedIn for large-scale content ranking



Key Metrics

Accuracy

CTR, NDCG@K, Recall@K

Fairness

Demographic Parity, Equalized Opportunity, GINI

Explainability

SHAP, Attention Heatmaps

Privacy

epsilon-DP, Federated Learning

Diversity

ILD, Coverage, Serendipity

Supporting Research

Federated Learning-Based Personalized Recommendation Systems: An Overview

IEEE Transactions on Consumer Electronics · 2023 · doi.org/10.1109/TCE.2023.3318754

Why it matters: This IEEE paper specifically studies how to add privacy protection (Federated Learning) to recommendation systems exactly like the Two-Tower model described on this slide.

What it found: It proves that training the model on each user's device separately — instead of sending all data to one central server — keeps user data private while losing only a small amount of accuracy. The raw data (your scroll patterns, your likes) never leaves your phone.

04 | Ethical Risks Identified

Bias & Discrimination

- Demographic bias: race/gender underrepresented in trends
- Political bias: amplification of ideologically extreme content
- Popularity bias: viral posts crowd out quality niche content

Privacy Violations

- Micro-profiling from scroll patterns, dwell time, mouse hover
- Cross-platform data linkage re-identifies anonymous users
- Shadow profiles built without user knowledge or consent

Filter Bubbles & Echo Chambers

- Engagement-maximising feeds reinforce existing beliefs
- Reduces exposure to diverse viewpoints and perspectives
- Documented radicalisation pathways (YouTube, Facebook)

Lack of Transparency

- Users cannot understand why content is shown or hidden
- Opaque content moderation inconsistently applied
- GDPR Article 22: right to explanation not fulfilled

Supporting Research

Artificial Intelligence and Algorithmic Bias? Field Tests on Social Networks with Teens

Technological Forecasting & Social Change · 2024 · ScienceDirect — doi.org/10.1016/j.techfore.2023.123076

Why it matters: This is one of the only papers that actually ran live experiments on real social media platforms to prove all four risks on this slide exist in practice — not just in theory.

What it found: Found real evidence of demographic bias (Black creators suppressed on TikTok), political bias (Twitter amplified extremes), privacy violations (detailed user profiling from micro-behaviours), and transparency failures where content moderation rules were applied inconsistently with no explanation to users.

Fairness in Recommender Systems: Research Landscape and Future Directions

User Modeling and User-Adapted Interaction · 2023 · doi.org/10.1007/s11257-023-09364-z

Why it matters: This research paper specifically studied filter bubbles and echo chambers in recommendation systems — exactly the bottom-left risk box on this slide.

What it found: Proved that filter bubbles are not an accident — they are a predictable mathematical result of optimising for engagement. The more the algorithm tries to keep you watching, the more it traps you in a bubble of the same opinions. Documented this on YouTube and Facebook.

05 | Responsible AI Techniques Proposed

Fairness Metrics

- Demographic Parity across user demographic groups
- Equalized Opportunity (EO) for content exposure rates
- GINI Coefficient for recommendation diversity
- Counterfactual Fairness via do-calculus interventions

Bias Mitigation

- Adversarial debiasing: penalise demographic-predictive features
- Post-hoc re-ranking: enforce group representation in Top-K
- Calibrated diversity regularisation during training
- Data augmentation for underrepresented content groups

Explainability Tools

- SHAP values: per-user feature attribution for feed decisions
- Attention weight visualisation on Transformer ranker
- Counterfactual explanations: 'Why not show X instead?'
- Global surrogate models (LIME) for policy-level audits

Privacy Techniques

- Differential Privacy (epsilon-DP) on gradient updates
- Federated Learning: train on-device, share only gradients
- K-anonymity grouping for user interest profiles
- Secure Aggregation: cryptographic gradient aggregation

Supporting Research

A Survey on Fairness-Aware Recommender Systems

Information Fusion (IEEE ICDM Best Student Paper) · 2023 · ScienceDirect — doi.org/10.1016/j.inffus.2023

Why it matters: This award-winning paper reviewed hundreds of studies to figure out which fairness measurement methods actually work best — directly validating the three metrics this slide proposes: Demographic Parity, Equalized Opportunity, and GINI coefficient.

What it found: It found that you need to measure fairness from two sides: the user's side (are all users getting equal quality recommendations?) AND the content creator's side (is all content getting equal chances of being seen?). Most systems only check one side.

Reducing Incidence of Biased Algorithmic Decisions Through Feature Importance Transparency

European Journal of Information Systems · 2024 · doi.org/10.1080/0960085X.2024.2395531

Why it matters: This paper directly tested whether using SHAP (the explainability tool on this slide) actually reduces bias in practice — or whether it is just a theoretical nice-to-have.

What it found: Good news: it works. When you show decision-makers which features drove an algorithmic decision (using SHAP), they are much more likely to catch and correct discriminatory patterns. Transparency is not just ethical window-dressing — it actively reduces bias.

Systematic Literature Review on Bias Mitigation in Generative AI

Springer AI & Ethics · 2025 · doi.org/10.1007/s43681-025-00721-9

Why it matters: This is the most recent (2025) academic review of every known bias-fixing technique — covering adversarial debiasing, data augmentation, and re-ranking, which are exactly the three methods proposed in the bias mitigation box on this slide.

What it found: After reviewing papers from 2020-2024, it concluded that adversarial debiasing combined with post-hoc re-ranking gives the best results. Neither alone is sufficient — you need to fix bias during training AND after training for best outcomes.

06 | Dataset & Analysis Plan

Twitter Sentiment & Engagement Dataset (Kaggle)

- ~1.6M labeled tweets (pos/neg/neutral)
- User IDs — engagement: likes, retweets, replies
- Hashtags — topics, trending categories, timestamps
- Content — text, media type, URL presence
- Metadata — follower count, verified status
- Supplementary: Reddit r/news dataset (Pushshift API)

5-Step Analysis Plan

- Step 1 — EDA: Explore engagement patterns; identify demographic disparities
- Step 2 — Baseline RS: Build CF + content-based; benchmark NDCG, CTR, Recall@K
- Step 3 — Bias Audit: Measure Demographic Parity & EO gaps
- Step 4 — Fairness Fix: Re-ranking + adversarial debiasing; measure delta accuracy vs delta fairness
- Step 5 — XAI + DP: SHAP explanations + epsilon-DP experiments; privacy trade-off curve

Supporting Research

A Survey on Federated Learning Systems: Vision, Hype and Reality for Data Privacy

IEEE Transactions on Knowledge and Data Engineering · 2023 · doi.org/10.1109/TKDE.2021.3124599

Why it matters: This IEEE paper directly supports Step 5 of the analysis plan — the differential privacy experiments and the privacy trade-off curve that the project needs to plot.

What it found: It tested every major privacy technique and measured exactly how much accuracy you lose for each level of privacy you gain. The finding: epsilon between 1 and 5 gives you strong privacy while keeping the model accurate enough to be useful in a real app.

Differentially Private Federated Learning with Adaptive Noise Mechanism

IEEE Transactions on Information Forensics and Security · 2024 · IEEE Xplore — Vol. 19, 2024

Why it matters: This paper provides the exact technical method for adding differential privacy noise to gradients in federated learning — which is what PySyft implements in this project.

What it found: Instead of adding the same fixed amount of noise to every gradient update (which wastes accuracy), it adapts the noise level based on how sensitive each update is. Result: better privacy AND better accuracy than the standard approach.

07 | Preliminary Observations & Expected Trade-offs

Three Core Trade-offs

Trade-off	Baseline	With Responsible AI	Best Setting	Observation
Accuracy vs Fairness (NDCG@10)	88%	+DP: 82% +DP+EO: 75%	lambda=0.3 Pareto-optimal	Fairness re-ranking reduces NDCG ~3%
Accuracy vs Explainability (CTR Score)	SVD: 65% LightFM: 78%	2-Tower: 88%	SHAP post-hoc negligible latency	Deep models gain ~15% CTR but need explainability
Accuracy vs Privacy (Recall@10)	epsilon=0.1: 48% epsilon=1: 65%	FedLearn: 82%	epsilon=1-5 FedLearn preferred	DP at epsilon=0.1 degrades Recall ~20%

Supporting Research

Differentially Private Federated Learning with Adaptive Noise Mechanism

IEEE Transactions on Information Forensics and Security · 2024 · IEEE Xplore — Vol. 19, 2024

Why it matters: This paper directly explains the numbers shown in the Accuracy vs Privacy bar chart on this slide — why epsilon=0.1 kills accuracy and why Federated Learning does better.

What it found: It confirmed that epsilon=1 to 5 is the sweet spot where privacy is strong enough to protect users but the model is still accurate enough to be useful. Federated Learning outperforms centralised DP because it avoids adding noise to the most important gradient updates.

Reducing Incidence of Biased Algorithmic Decisions Through Feature Importance Transparency

European Journal of Information Systems · 2024 · doi.org/10.1080/0960085X.2024.2395531

Why it matters: This paper backs the middle bar chart showing that SHAP explainability barely hurts accuracy — in fact SVD at 65% vs Two-Tower at 88% is about model power, not explainability.

What it found: Proved experimentally that adding SHAP explanations on top of a complex model costs almost nothing in terms of speed or accuracy. You can have a powerful Two-Tower model AND full SHAP transparency at the same time — they do not conflict.

08 | Team Contributions & Next Steps

Member 1 — Data & Modeling

- Dataset acquisition & EDA (Twitter Kaggle)
- Build baseline RS: content-based + collaborative filtering
- Implement Two-Tower ranking model
- Run fairness audit: Demographic Parity, EO metrics

Member 2 — Ethics & Evaluation

- Apply SHAP for explainability & attention visualisation
- Implement bias mitigation: re-ranking + adversarial debiasing
- Run Differential Privacy & Federated Learning experiments
- Trade-off analysis, final report & presentation

Project Timeline

Week 1-2 [In Progress]	EDA + baseline RS training + initial bias audit
Week 3-4	Fairness metrics + re-ranking + adversarial debiasing
Week 5	SHAP explainability analysis + visualisation
Week 6	Differential Privacy + Federated Learning experiments
Week 7	Trade-off curves + comparative analysis + final report
Final Week	Presentation prep + live demo + Q&A

Full Reference List

All papers cited throughout this document:

- [IEEE Access \(2024\) — A Systematic Literature Review on AI-Based Recommendation Systems and Ethical Considerations](#)
- [IEEE ICDM / Information Fusion \(2023\) — A Survey on Fairness-Aware Recommender Systems](#)
- [Springer UMuAI \(2023\) — Fairness in Recommender Systems: Research Landscape and Future Directions](#)
- [Technological Forecasting & Social Change \(2024\) — AI and Algorithmic Bias: Field Tests on Social Networks](#)
- [European Journal of Information Systems \(2024\) — Reducing Biased Algorithmic Decisions Through Feature Importance Transparency](#)
- [IEEE Transactions on Consumer Electronics \(2023\) — Federated Learning-Based Personalized Recommendation Systems](#)
- [IEEE Transactions on Knowledge & Data Engineering \(2023\) — Survey on Federated Learning Systems](#)
- [IEEE Transactions on Information Forensics & Security \(2024\) — Differentially Private Federated Learning with Adaptive Noise](#)
- [Springer AI & Ethics \(2025\) — Systematic Literature Review on Bias Mitigation in Generative AI](#)

Thank You

Questions & Discussion

Core Insight: Social media recommendation systems are among the most ethically consequential AI deployments in existence. By rigorously applying fairness metrics, SHAP-based explainability, and differential privacy — and grounding each technique in peer-reviewed IEEE and Springer research — this project demonstrates that accuracy and responsibility are not mutually exclusive: they are complementary goals.

Dataset: Twitter Sentiment (Kaggle) · Tools: Python · Fairlearn · SHAP · PySyft

Research Papers: IEEE Access · IEEE TKDE · IEEE TCE · IEEE TIFS · Springer AI&Ethics · UMUAI